



SEEK-O-SPHERE

README

Project Overview

A free, interactive website that teaches Artificial Intelligence to everyone, ages 8 to 80. Plain HTML, CSS & JavaScript on a Firebase backend.

AI is a tool. You are the thinker. Eulid agrees.

What It Is

Seek-O-Sphere teaches Artificial Intelligence in plain language to learners of every age, pairing readable chapters with hands-on practice and personal progress tracking — built with vanilla web technology and Firebase, hosted free on GitHub Pages.

Tech Stack

Layer	Technology
Markup	HTML5 — single-page app (<code>index.html</code>) + auth/dashboard pages
Styling	Vanilla CSS, custom "Sienna" design system
Scripting	Vanilla JavaScript (ES6+), no framework
Fonts	Google Fonts — Nunito + Inter
Questions	Local engine (algorithmic + JSON), optional Claude API
Auth	Firebase Authentication (compat v10.12.0)
Database	Cloud Firestore (compat v10.12.0, offline persistence)
Hosting	GitHub Pages

What the Site Includes

- **Home** — hero, the Euclid mascot (eyes follow the cursor), animated neural-network canvas, seamless brand marquee.
- **Learn chapters** — What is AI?, History, Types, Study Tools, Using AI Responsibly.
- **History (interactive)** — 1950→2026 year scrubber, era ribbon, "Then/Now" focus card, landmark chips, "Guess the Year".
- **Visual Learning Lab** — Curriculum Lab (CBSE/ICSE/UK/US/IB · Math, Science, English, History, Geography) and Exam Prep (CUET, Railways, Bank PO, SSC, UPSC), with tutor, hints, explanations, streaks and saved progress.
- **Find My AI Tool** — a keyword-scored recommender over 20+ tools.
- **Help Centre** — searchable FAQ with a topic-card landing.
- **Accounts** — email/Google sign-in plus guest browsing.
- **Student Dashboard** — profile picture, accuracy/streak/active-days, progress chart, per-subject breakdown, heatmap, sessions.
- **Admin Panel** — all users with scores, per-user progress viewer, role management.

Project Structure

```
ai-classroom/
├─ index.html           main single-page app (all learning sections)
├─ README.md
├─ BACKEND_SETUP.md
├─ firestore.rules      security rules (publish in Firebase console)
├─ favicon.svg
├─ pages/               login · dashboard · admin · account-settings · content pages
├─ css/                 base/ · components/ · layout/ · pages/ · dashboard.css
├─ js/                  session.js · questions-engine.js · core/ · components/ · db/ · auth/
├─ data/                question-bank.json · tools-data.js
├─ assets/images/       page illustrations
```

Two faces. `index.html` is the canonical experience (sections shown via `showPage()`). `pages/` holds the authenticated flows plus standalone content pages sharing `js/core/nav.js`. Live progress-saving runs inline in `index.html`; `js/db/` and `js/components/` are supporting/legacy modules.

Design System (Sienna)

Token	Value	Role
--brand	#8B3620	Primary sienna
--brand-dark	#46200F	Nav / dark backgrounds
--accent	#C0562F	Terracotta CTAs
--green	#5F7355	Correct answers / success
--text	#3E2B1E	Body text
--muted	#8B6E5A	Secondary text

Headings in Nunito (700–900), body in Inter (400–600). Mascot: **Eulid**. Wordmark: "Seek-O-Sphere", the **O** a hand-built solar-system SVG.

Local Development

Fully static — open with any static server (VS Code Live Server used here). No build step, no `npm install`.

```
Home:      http://127.0.0.1:5501/ai-classroom/index.html
Login:     http://127.0.0.1:5501/ai-classroom/pages/login.html
Dashboard: http://127.0.0.1:5501/ai-classroom/pages/dashboard.html
Admin:     http://127.0.0.1:5501/ai-classroom/pages/admin.html
```

Deployment

```
git checkout dev && git add . && git commit -m "message" && git push origin dev
git checkout main && git merge dev && git push origin main && git checkout dev
```

Publish the rules. GitHub Pages serves static files only — it does not apply `firestore.rules`. Publish them in the Firebase console (see `BACKEND_SETUP.md`).

Backend at a Glance

Firebase project `ai-classroom-ad779` (Firestore region `asia-south2`). Auth: Email/Password, Google, Anonymous. SDK: compat v10.12.0 via CDN. Offline persistence on.

```
users/{uid}
  name, email, role, curriculum, ageGroup, photo,
  totalQuestions, correctTotal, bestStreak, createdAt, lastActive

progress/{uid}/daily/{YYYY-MM-DD__mode__subject}
  date, mode, subject, label, questions, correct, lastTs
```

Question Engine

```
const QUESTION_SOURCE = 'json'; // 'json' = free local · 'api' = live Claude
```

json: algorithmic math + curated JSON bank (free, offline). **api**: Claude (`claude-sonnet-4-6`) via a Cloudflare Worker proxy, with the JSON bank as fallback.

Making Yourself an Admin

1. Sign up on the site.
2. Firebase console → Firestore → `users`.
3. Open your document (ID = your `uid`).
4. Change `role` from `student` to `admin`.
5. Sign out and back in — you'll land on the Admin Panel.

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